V-JEPA 2.1 on IntPhys 2 (Main split)

Table 1: Accuracy values showing best model performance for each difficulty level. Rows other than V-JEPA 2.1 are reproduced from IntPhys 2 (Bordes et al., 2025), Table 2: each model's best run per column over about a dozen hyperparameter sets. For V-JEPA 2.1 (shaded, ours), each column is the best of the 6 context lengths, with surprise aggregated over time by its maximum (protocol of the released code) or its mean. Bold: best per model type, ours included.

Model	Type	Easy	Medium	Hard	Overall	Held Out	*IntPhys*
GPT4-o	MLLMs	57.69	54.75	54.17	53.75	53.19	-
Qwen-VL 2.5	MLLMs	50.96	53.25	51.49	52.27	49.12	-
Gemini-1.5 Pro	MLLMs	58.65	53.0	52.67	52.27	52.10	*55.81*
Gemini-2.5 Flash	MLLMs	**64.42**	**56.75**	**54.46**	**55.63**	**56.10**	*56.39*
VideoMAEv2-g	Predictive	46.00	**58.50**	52.73	53.75	53.49	*59.40*
Cosmos-4B	Predictive	46.00	52.00	48.05	49.41	48.84	*85.42*
V-JEPA-h + RoPE	Predictive	52.00	53.00	57.42	53.75	54.65	*98.30*
V-JEPA 2-h	Predictive	54.00	**58.50**	**59.38**	**57.51**	**56.40**	*87.22*
V-JEPA 2.1-g, max surprise	Predictive	50.00	52.00	54.76	51.78	–	-
V-JEPA 2.1-g, mean surprise	Predictive	**63.46**	55.50	55.95	55.14	–	-
Human	-	96.17	97.8	95.5	96.44	92.44	-

Table 2: Accuracy across property and camera type. For each model, the accuracy of each subset is the best one across a set of hyperparameters (for V-JEPA 2.1: the 6 context lengths). The smaller size of each subset contributes to volatility in performance (52 / 68 pairs per Fixed / Moving cell, 94 for Solidity Moving).

	Permanence		Immutability		Continuity		Solidity	
Model	Fixed	Moving	Fixed	Moving	Fixed	Moving	Fixed	Moving
GPT4-o	59.62	**58.82**	58.65	59.56	**54.81**	**57.35**	**56.73**	55.32
Qwen-VL 2.5	53.85	54.41	56.73	53.68	52.88	54.41	50.96	51.06
Gemini-1.5 Pro	55.77	55.88	56.73	56.73	54.80	54.80	56.73	56.73
Gemini-2.5 Flash	**64.42**	**58.82**	**59.62**	**63.97**	**54.81**	55.15	55.77	**56.38**
VideoMAEv2-g	63.46	50.00	54.81	53.69	**65.38**	54.41	48.08	**59.57**
Cosmos-4B	51.92	41.18	50.96	48.32	53.85	50.00	48.08	55.32
V-JEPA-h + RoPE	63.46	**67.65**	51.92	56.38	50.00	57.35	50.00	52.13
V-JEPA 2-h	59.62	57.35	**55.77**	**58.72**	57.69	**75.00**	46.15	58.51
V-JEPA 2.1-g, max surprise	44.23	55.88	51.92	50.00	59.62	52.94	51.92	53.19
V-JEPA 2.1-g, mean surprise	**67.31**	54.41	**55.77**	54.41	59.62	60.29	**53.85**	55.32
Human	100.0	99.26	97.11	90.44	99.04	94.44	96.15	95.21

Experimental setup. *Model.* Official V-JEPA 2.1 ViT-g/16 release at 384 px from facebookresearch/vjepa2 (checkpoint `vjepa2_1_vitg_384.pt`). The model code is vendored unchanged from that repository, and the context encoder, target (EMA) encoder (1.01 B parameters each) and predictor (55 M) load with strict key matching. V-JEPA 2.1 predicts the concatenation of 4 hierarchical encoder levels; we use only the last one (layer 39), layer-normed as in its training targets, and the predictor's mask token 0 (the one used for video during training).

Surprise. Following Garrido et al. (2025) and the setup reported as best for V-JEPA 2 in IntPhys 2: videos are sampled every 10th frame (6 fps), split into 48-frame windows with stride 2, and for each window the model predicts the features of all frames after a context of $C \in \{12, 18, 24, 30, 36, 42\}$ frames (the paper's $\{4, \ldots, 14\}$ scaled by 48/16). The prediction error is the L1 distance between predicted and target features (bf16). With `max_context_mode`, the first window is also predicted from every shorter context $2, 4, \ldots, C-2$, in case the physics-breaking event happens early. A video's surprise is the maximum (or mean) of these errors over time.

Metric. Relative accuracy on matched pairs: in each scene quadruplet, the impossible video `k_Impossible` must be more surprising than its matched `k_Possible` ($k = 1, 2$), giving 506 pairs on Main (1,012 videos), with 95% confidence intervals of about ±4.4 points overall and ±14 on Easy. Easy/Medium/Hard follow the environment lists of the authors' scoring notebook (52/200/168 pairs; the other 86 pairs count only towards Overall). The paper's Easy and Hard values are not multiples of 1/52 and 1/168, so its subsets differ slightly. Held Out was not evaluated.

Observation. For 81–97% of videos, the maximum error comes from the short-context predictions on the first window rather than the sliding windows, so under max surprise the context length barely matters, and accuracy stays at chance. The mean surprise is modestly above chance.

Compute. 6 single-GPU nodes (RTX 5090, 3090, 5070 Ti, 5060 Ti, 4060 Ti, plus one more 16 GB card), ≈21 h wall-clock, 1.3–7.3 min per video depending on the GPU; videos split evenly across ranks, with no inter-node communication.